Geometry Under Constraints: Projections, Packing, and Symmetric Coordinates

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A geometric extremum depends as much on the feasible set as on the formula being optimized. Squared distances invite orthogonal projection; circle packing requires distance constraints between centers; symmetric rational equations become simpler after removing an irrelevant common scale. Each situation suggests a different set of coordinates.

§1.1 A distance problem is a projection problem

Let $A = (0, -2)$, $B = (2, -2)$, and let $M = (x, y)$ lie on $x + y = 1$. We want to minimize

$$|M - A|^2 + |M - B|^2.$$

Put $C = (A + B)/2 = (1, -2)$. Expanding inner products gives the exact identity

$$\boxed{|M - A|^2 + |M - B|^2 = 2|M - C|^2 + \frac{1}{2}|A - B|^2.}$$

It holds in any inner-product space, not just in the plane. The distance $|A - B| = 2$ is fixed, so minimizing the sum is equivalent to projecting C orthogonally onto the constraint line.

The line has normal vector $n = (1, 1)$. Its signed equation is $n \cdot M = 1$, while $n \cdot C = -1$. Therefore the projection is

$$M_* = C + \frac{1 - n \cdot C}{\|n\|^2} n = (1, -2) + (1, 1) = (2, -1).$$

The squared distance from C to the line equals 2. Hence

$$\min_{x+y=1} (|M - A|^2 + |M - B|^2) = 2 \cdot 2 + 2 = 6.$$

There is no maximum because the constraint line is unbounded and the squared distance grows quadratically along it.

This same identity underlies least squares: minimizing sums of squared distances amounts to projecting onto an affine subspace, with the residual orthogonal to all allowed directions.

§1.2 The Parabola's Farthest Point

Take a downward-opening parabola

$$y = ax^2 + bx, \qquad a < 0,\ b > 0,$$

restricted to the interval between its two x-intercepts, $0 \le x \le -b/a$. The squared distance to the origin is

$$F(x) = x^2 + (ax^2 + bx)^2, \qquad F'(x) = 2x + 2(ax^2 + bx)(2ax + b).$$

Although the height of the parabola is maximal at $x = -b/(2a)$, the distance to the origin also depends on x^2. The vertex therefore need not maximize F.

For the concrete parabola $y = 2x - x^2$ on $[0, 2]$,

$$F(x) = x^2 + x^2(2 - x)^2, \qquad F'(x) = 2x(2x^2 - 6x + 5).$$

The quadratic factor has discriminant $36 - 40 = -4$ and positive leading coefficient, so $F'(x) > 0$ for $0 < x \leq 2$. Thus the farthest point is $(2, 0)$, at squared distance 4, while the vertex $(1, 1)$ has squared distance only 2.

An extremum of one coordinate is not generally an extremum of a nonlinear function of all coordinates. The correct objective must be differentiated on the actual feasible domain.

§1.3 Packing circles becomes a configuration-space problem

Suppose n closed disks of radius $1/2$ must fit inside a container $C \subseteq \mathbb{R}^2$. Instead of manipulating the disks directly, represent each by its center c_i. The conditions are

$$\|c_i - c_j\| \geq 1 \quad (i \neq j), \qquad c_i \in C \ominus B_{1/2},$$

where $C \ominus B_{1/2}$ is the set of points whose radius-1/2 disk is wholly contained in C. This operation is called morphological erosion or Minkowski subtraction.

For two unit-diameter disks inside a round container of radius R, both centers must lie within radius $R - \frac{1}{2}$ of its center. Their greatest possible separation is $2R - 1$, which must be at least 1. Thus $R \geq 1$, and placing the disk centers at opposite ends of a diameter attains equality. The minimal container diameter is exactly 2.

For two disks inside a square of side s, each center must lie in the smaller square of side $s - 1$. The largest possible distance between two such centers is $\sqrt{2}(s - 1)$. Hence

$$\sqrt{2}(s - 1) \geq 1 \quad \Longleftrightarrow \quad \boxed{s \geq 1 + \frac{1}{\sqrt{2}}.}$$

Opposite corners of the allowed center square achieve the bound.

Already for $n = 2$, the exact results are much stronger than the area comparison $n\pi/4 \leq \text{area}(C)$. For larger n, the feasible configurations form a constrained subset of C^n from which all pairwise collision regions have been removed. This configuration-space geometry, rather than area alone, explains the difficulty of optimal packings.

§1.4 Three equal disks force an equilateral configuration

The two-disk examples reveal the constraints, but a third disk makes the geometry more interesting. What is the smallest circular container that can hold three disks of radius $1/2$ without overlap? A naive area estimate would only give

$$\pi R^2 \geq 3\pi(1/2)^2, \qquad R \geq \frac{\sqrt{3}}{2},$$

which is far from sharp because it ignores how centers are separated.

Write c_1, c_2, c_3 for the disk centers, measured from the container center, and let

$$r = R - \frac{1}{2}.$$

Containment implies $\|c_i\| \le r$, while nonoverlap implies $\|c_i - c_j\| \ge 1$ for each pair $i \ne j$. The key is a simple identity:

$$\boxed{\sum_{i<j} \|c_i - c_j\|^2 = 3\sum_{i=1}^{3} \|c_i\|^2 - \left\|\sum_{i=1}^{3} c_i\right\|^2.}$$

Expand the squares to verify it; each $\|c_i\|^2$ occurs twice on the left, and each pairwise dot product occurs with coefficient -2. The right-hand side rearranges exactly those terms.

The three separation constraints give a lower bound of 3 for the left side. For the right side,

$$3 \le \sum_{i<j} \|c_i - c_j\|^2 \le 3\sum_{i} \|c_i\|^2 \le 9r^2.$$

Therefore $r \ge 1/\sqrt{3}$, or

$$\boxed{R \ge \frac{1}{2} + \frac{1}{\sqrt{3}}.}$$

This lower bound is achieved. Place the three centers at the vertices of an equilateral triangle of side 1, centered at the origin. Each center lies at distance $1/\sqrt{3}$ from the origin, so the outer container of radius $1/2+1/\sqrt{3}$ touches all three disks while the disks touch one another.

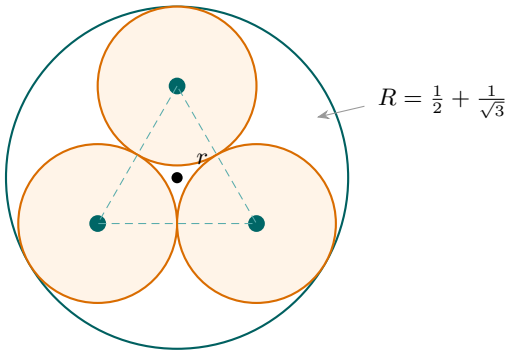


Figure 1.1: The sharp three-disk packing. The dashed equilateral triangle has side length 1 and circumradius $1/\sqrt{3}$; the disks have radius $1/2$.

The inequality is stronger than an area bound because it keeps a *pairwise distance invariant*. Equality in the algebra forces all centers onto the allowed circle, their vector sum to vanish, and all pairwise separations to equal one. Thus the symmetric arrangement is not merely a plausible picture; the calculation explains the rigidity of the optimizer.

§1.5 Symmetric rational equations live on projective ratios

Consider positive a, b, c satisfying

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = 4.$$

If (a, b, c) solves the equation, so does (ta, tb, tc) for every $t > 0$. Hence the equation constrains only the ratios among a, b, c. It is natural to introduce the elementary symmetric quantities

$$s = a + b + c, \qquad p = ab + bc + ca, \qquad q = abc.$$

Because $\frac{a}{b+c} = \frac{s}{s-a} - 1$, the original equation becomes

$$s\left(\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c}\right) = 7.$$

The pairwise products of the denominators sum to $s^2 + p$, while their product equals $sp - q$. Clearing the strictly positive denominator yields the homogeneous cubic

$$\boxed{s^3 - 6sp + 7q = 0.}$$

There is an actual family of positive solutions, not just a formal projective equation. Fix the overall scale by taking $b = c = 1$ and write $a = t > 0$. The original condition becomes

$$\frac{t}{2} + \frac{2}{t+1} = 4.$$

Clearing the positive denominator gives

$$t(t+1) + 4 = 8(t+1), \qquad t^2 - 7t - 4 = 0.$$

The unique positive solution on this symmetric ray is

$$\boxed{t = \frac{7 + \sqrt{65}}{2}.}$$

The other quadratic root is negative. Thus all positive triples proportional to

$$\left(\frac{7 + \sqrt{65}}{2},\ 1,\ 1\right)$$

satisfy the rational equation. The cubic derived in s, p, q gives the same result: when $s = t + 2,\ p = 2t + 1,\ q = t$, its left side factors as

$$s^3 - 6sp + 7q = (t+1)(t^2 - 7t - 4).$$

The factor $t+1$ is never zero in the positive region. This explicit section of the projective solution set illustrates why removing scale helps: one homogeneous equation in three variables becomes a scalar quadratic after intersecting with a geometrically meaningful symmetry ray.

This equation defines a projective cubic locus in the ratios $[a : b : c]$, restricted to the positive region. An elementary inequality still gives a useful geometric boundary: by Nesbitt's inequality,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2},$$

with equality at $a = b = c$. The level 4 therefore lies well away from the balanced point.

The projection calculation, the disk-center constraints, and the homogeneous cubic all remove irrelevant coordinates. For squared distances we eliminate translation using a midpoint; for packing we replace congruent disks by their centers; for a symmetric rational equation we quotient out overall scale. In each case the improved coordinates reveal the geometry of the feasible set before optimization begins.